

OPERATIONAL AMPLIFIER AND APPLICATIONS

An operational amplifier or op-amp is a direct coupled multistage voltage with extremely high gain. Its behaviour can be controlled by adding suitable feedback. It has a very high input impedance and very low output impedance. Earlier day's op-amp was used for performing mathematical operations such as addition, subtraction, multiplication, integration and differentiation. Hence the device acquired the name operational amplifier.

Circuit symbol for an Op-amp.

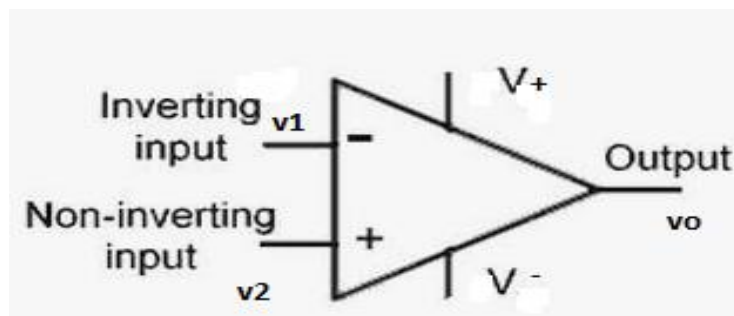


Figure 1. Circuit symbol for an op-amp.

There are 5 terminals namely:

1. A non-inverting input terminal denoted by '+' symbol.
2. An inverting input terminal denoted by '-' symbol.
3. An output terminal.
4. A positive supply voltage terminal denoted by 'V+'.
5. A negative supply voltage terminal denoted by 'V-'.

From the above figure;

V_1 = voltage at the inverting input.

V_2 = voltage at the non inverting input.

V_o = output voltage.

All the voltages are measured w.r.t ground.

Supply voltages denoted by V_+ and V_- and are in the range $\pm 9V$ to $\pm 22V$.

A voltage applied to non-inverting input produces an **in phase** or same polarity voltage at the output while a voltage applied to inverting input produces an out of phase or opposite polarity voltage at the output.

The output voltage denoted by V_o is proportional to difference between the input voltages.

Hence, $v_o = A(v_2 - v_1)$, where, A is the voltage gain.

Hence op-amp is basically a differential or difference amplifier. The op-amp is a differential amplifier which amplifies the difference between its two input voltages. Hence,

$v_o = A(v_2 - v_1) = Av_d$ where, $v_d = (v_2 - v_1)$ is differential or difference input voltage, and A = open loop voltage gain.

When a voltage v_1 is applied to inverting input terminal, with non inverting input terminal grounded, ($v_2 = 0$), the output voltage is $v_o = A(v_2 - v_1) = A(0 - v_1) = -Av_1$.

This indicates that output voltage will be inverted (phase or polarity reversed) w.r.t the applied input voltage.

On the other hand, when a voltage v_2 is applied to non-inverting input terminal with the inverting input terminal grounded ($v_1 = 0$), the output is $v_o = A(v_2 - v_1) = A(v_2 - 0) = Av_2$ which indicates that the output voltage will have the same phase or polarity as input voltage.

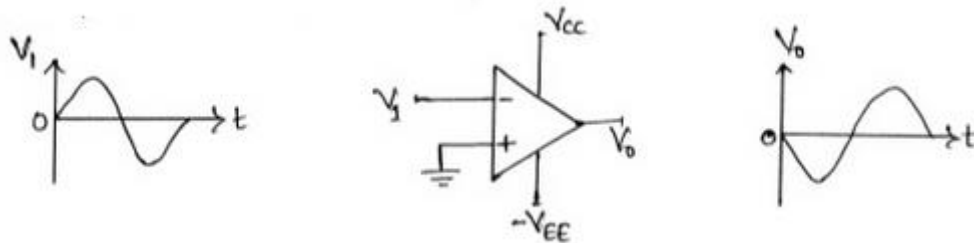


Figure 2. Inverting Amplifier

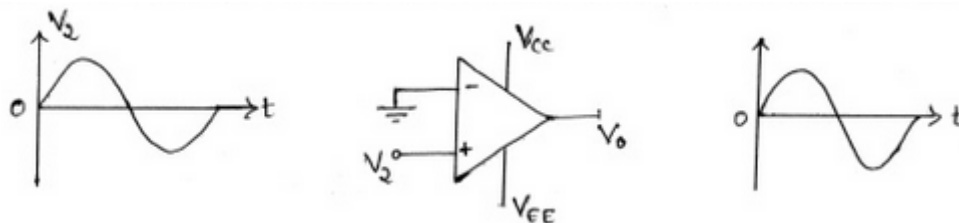


Figure 3. Non-inverting Amplifier

Characteristics of Ideal Op-amp

1. An ideal op-amp has Infinite input resistance ($R_i = \infty$).
2. An ideal op-amp has zero output resistance ($R_o = 0$).
3. Infinite voltage gain (i.e., $A = \infty$)
4. An ideal op-amp amplifies signals of any frequency with a constant gain, which means it has infinite bandwidth ($B.W. = \infty$)

5. When equal voltages are applied to both inputs the output voltage is zero. Thus it has perfect balance.
6. Common mode rejection ratio is infinite (*i.e.*, $CMRR = \infty$)
7. An ideal op-amp has infinite slew rate (*i.e.*, $S = \infty$)
8. The characteristics an ideal op amp does not change with temperature.

Characteristics of a practical op amp

1. Open loop voltage gain is not infinite, but generally in the range of 10^4 to 10^6 or more.
2. Input resistance is not infinite, but it is high in the order of mega ohms.
3. Output resistance is not zero but in the range of $100\ \Omega$ or less.
4. Bandwidth is not infinite but in the range of 100 MHz.
5. The Common mode rejection ratio is not infinite, but in the order of 90 dB. Negative and positive voltage swings (output voltages) are limited by supply voltage, V_+ and V_- .

Typical parameter values for $\mu A741$ IC

1. Open loop voltage gain, $A = 2 \times 10^5$
2. Input resistance, $R_i = 2M\Omega$
3. Output resistance, $R_o = 75\Omega$
4. CMRR=90 dB
5. Band width, BW=1M Hz.
6. Slew rate=0.5V/ μ s

Pin configuration of op-amp $\mu A741$ IC

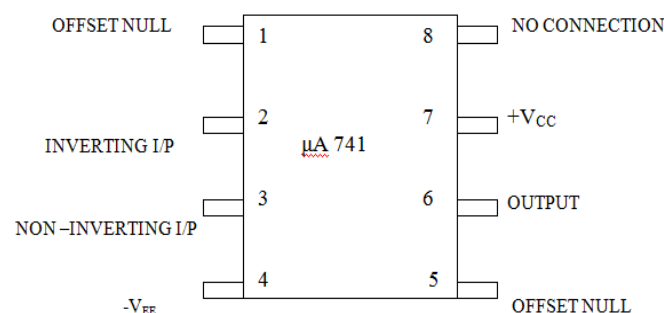


Figure 4. Pin configuration

Offset voltage: It is the output voltage that occurs even when both the input voltages are zero.

Equivalent circuit of an op-amp

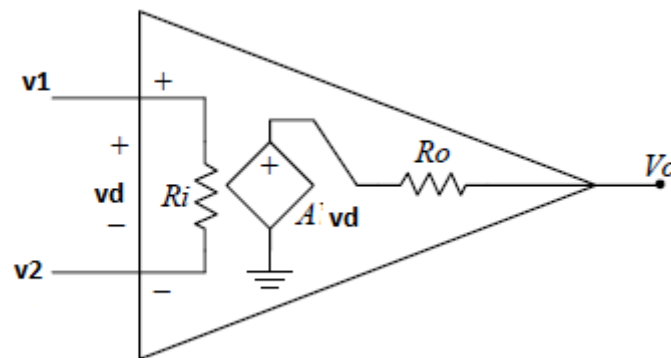


Figure 5. Equivalent circuit

Figure 5 shows Equivalent circuit of an op-amp, in which R_i represents input resistance measured between input terminals. It is also called differential input resistance.

Differential input voltage, $v_d = (v_2 - v_1)$ is multiplied by open loop voltage gain A and appears as Av_d at the output. R_o represents the output resistance.

Saturable property of an op amp

The property by which Op-amp output saturates at two saturation levels ($\pm V_{sat}$) decided by the supply voltages is called as saturable property of Op-amp.

As the open loop gain of op-amp is very large, of the order of 10^5 or more, even a very small difference input voltage ($v_2 - v_1$) produces extremely high output voltage.

However maximum output voltage is limited by supply voltage.

As a rule, maximum output voltage may be taken 1.5V less than supply voltage.

For example for a supply voltage of $\pm 15V$, the output voltage is limited to maximum of $\pm 13.5V$. Once the output reaches this limit, it does not increase further even if the magnitude of input voltage is increased. Under this condition op-amp is said to be clipped or saturated.

Example, $A=2 \times 10^5$ and saturation voltage $= \pm 13.5V$.

$$v_o = A(v_2 - v_1)$$

$$(v_2 - v_1) = \frac{v_o}{A} = 13.5 / 2 \times 10^5 = 67.5 \mu V$$

Therefore magnitude of differential input voltage that causes op-amp to saturate is $67.5 \mu V$.

For higher values of $(v_2 - v_1)$ output will be limited either to $+13.5V$ or $-13.5V$.

Slew rate (SR) of an op-amp

Slew rate is defined as maximum time rate of change of its output voltage, expressed in volts per microsecond.

$$\text{SR or Max. SR} = \left. \frac{dv_o}{dt} \right|_{\max} \text{ V}/\mu\text{s}.$$

Slew rate is a measure of how fast the op-amp output can change in response to changes in the input signal.

Differential gain

When two input voltages v_1 & v_2 such that $v_1 \neq v_2$ are applied to an op-amp, the output voltage is given by

$$v_o = Av_d = A_d v_d = A_d(v_2 - v_1)$$

where $A_d = A = \frac{v_o}{v_d}$ is called differential gain of op-amp.

Common-mode gain of op-amp

When two equal input voltages are applied to op-amp i.e., $v_1 = v_2 = v_c$, then the output voltage must be ideally zero.

However, in a practical op-amp a small non-zero output voltage exists that is given by

$$v_o = A_c v_c$$

where $A_c = \frac{v_o}{v_c}$ is called common mode gain of the op-amp.

Op amp parameters

- **Input offset voltage (V_{io});** Input offset voltage is the voltage that must be applied between the two input terminals such that the output voltage becomes zero. Typical values of $V_{io} = 1 \text{ mV}$.
- **Input offset current (I_{io});** It is the algebraic difference between the currents flowing into non-inverting and inverting terminals.
i.e., Input offset current $I_{io} = |I_{B1} - I_{B2}|$
where I_{B1} = current into inverting terminal. I_{B2} = current into non-inverting terminal
- **Common mode rejection ratio (CMRR):** CMRR is defined as the ratio of differential gain A_d to the common mode gain. $\text{CMRR} = \rho = \frac{|A_d|}{|A_c|}$

CMRR is often expressed in decibels as $CMRR = 20\log_{10}\frac{|A_d|}{|A_c|}$ dB. Typical value of CMRR for $\mu A741$ IC is 90dB. For an ideal op amp A_d infinite and A_c is zero so that CMRR is infinite. For a practical op amp A_d is very large A_c is non zero but small. So that CMRR is very large. CMRR is a measure of the op-amp to reject signals common to both inputs.

- **Concept of virtual short in an op-amp**

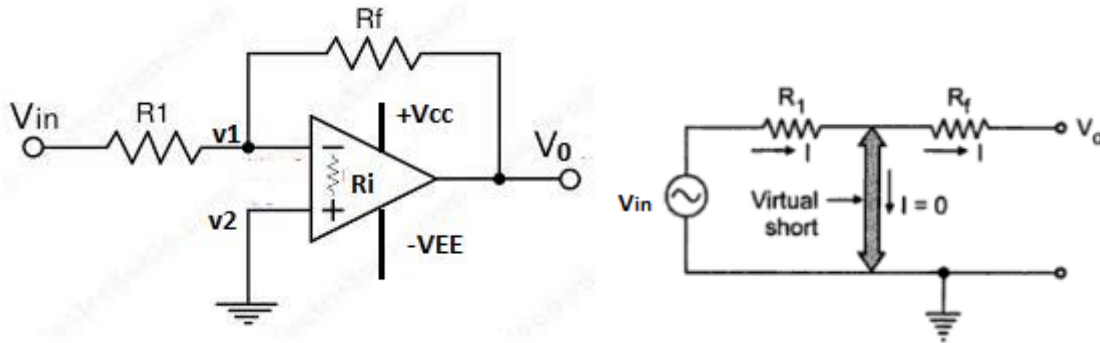


Figure 6. Virtual ground concept

Figure 6 shows circuit of op-amp inverting amplifier with negative feedback.

R_i represents input resistance. Output voltage v_o is given by

$$v_o = A (v_2 - v_1) \text{ or } (v_2 - v_1) = \frac{v_o}{A}$$

where A is open loop voltage gain of op-amp. The output voltage v_o cannot exceed the DC supply voltage given to the op-amp. For $\mu A741$ IC, supply voltage is 12V and open loop voltage gain is 2×10^5 . To get an output voltage of 10V by applying an input voltage of 1 V, the required differential input voltage is, $(v_2 - v_1) = \frac{v_o}{A} = 10V / 2 \times 10^5 = 50\mu V$. This value is very small compared to input & output voltages and may be considered as 0V.

$$\text{i.e., } (v_2 - v_1) \cong 0V \text{ or } v_2 = v_1$$

Above equation indicates that inverting and non-inverting input terminals are at same potential. Therefore voltage across R_i is zero. No current flows from input terminals to ground. The virtual short is also called as virtual ground.

OP-Amp Applications

Op-amp Inverting Amplifier

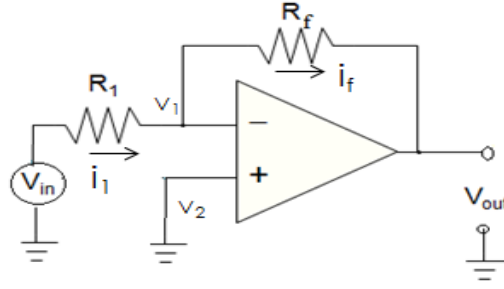


Figure 7. Inverting amplifier

Input signal v_{in} is applied to inverting input terminal and non-inverting input terminal is grounded. Feedback from the output to inverting input terminal is provided through the feedback resistor R_f .

Since non inverting input terminal is grounded, $v_2 = 0$. Due to virtual ground at the input of op-amp, $v_1 = v_2 = 0$

(1)

Due to high input impedance, current flowing into its inverting input terminal is zero.

Same current flows through R_1 and R_f .

$$\text{i.e., } i_1 = i_f \quad (2)$$

$$\text{But, } i_1 = \frac{v_{in} - v_1}{R_1}$$

Since, $v_1 = 0$,

$$i_1 = \frac{v_{in}}{R_1} \quad (3)$$

$$i_f = \frac{v_1 - v_{out}}{R_f} = \frac{-v_{out}}{R_f} \quad (4)$$

Sub (3) & (4) in (2) we get,

$$\frac{-v_{out}}{R_f} = \frac{v_{in}}{R_1}$$

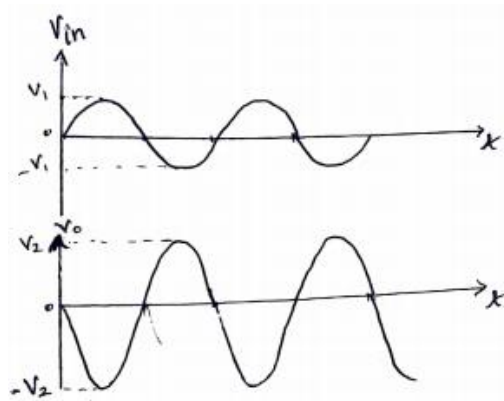
Or, output voltage of an inverting op-amp is $v_{out} = -\left(\frac{R_f}{R_1}\right) v_{in}$

The closed loop gain is

$$A_f = \frac{v_{out}}{v_{in}} = \frac{-R_f}{R_1}$$

where A_f is closed loop gain with negative feedback.

Input/output waveforms:



Op amp Non-Inverting Amplifier

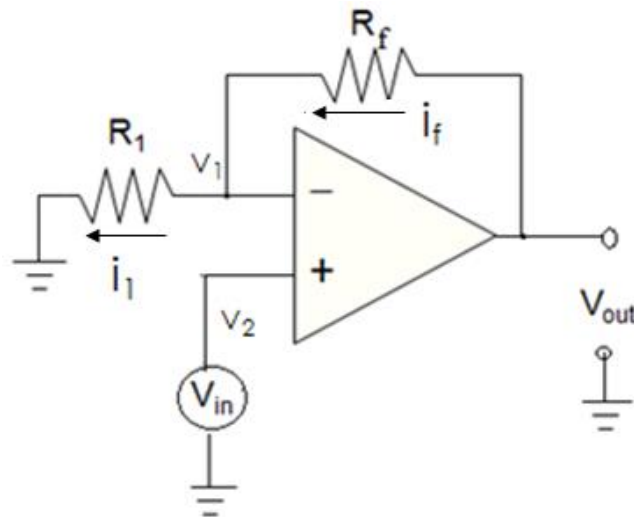


Figure 8. Non-Inverting amplifier

Here, input voltage is applied to non-inverting terminal of op-amp and inverting terminal grounded. Feedback resistor R_f is connected to inverting terminal. Due to virtual ground at the input terminals of op-amp, $v_1 = v_2 = 0$

Since, $v_2 = v_{in}$, $v_1 = v_{in}$

$$\therefore v_1 = v_2 = v_{in} \quad (1)$$

Due to high input impedance, current does not flow into the input terminals of the op-amp. Hence, $i_1 = i_f$ (2)

$$\text{But, } i_1 = \frac{v_1}{R_1} = \frac{v_{in}}{R_1} \quad (3)$$

$$i_f = \frac{v_{out} - v_1}{R_f} = \frac{v_{out} - v_{in}}{R_f} \quad (4)$$

Sub (3)&(4) in (2),

$$\frac{v_{in}}{R_1} = \frac{v_{out} - v_{in}}{R_f}$$

$$\frac{v_{out}}{R_f} = \frac{v_{in}}{R_f} + \frac{v_{in}}{R_1}$$

$$\frac{v_{out}}{R_f} = v_{in} \left(\frac{1}{R_f} + \frac{1}{R_1} \right)$$

$$\frac{v_{out}}{v_{in}} = R_f \left(\frac{1}{R_f} + \frac{1}{R_1} \right)$$

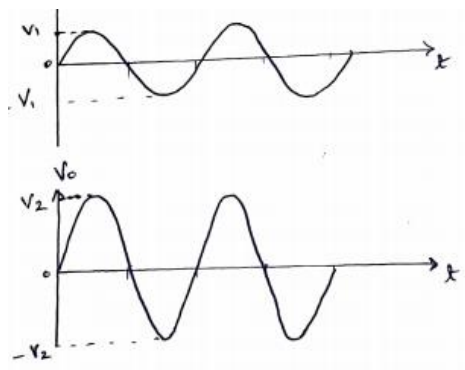
The output voltage of a non-inverting op-amp is

$$V_{out} = V_{in} \left(1 + \frac{R_f}{R_1} \right)$$

The closed loop gain is, $A_f = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_1}$

where A_f is closed loop gain with negative feedback.

Input/output waveforms



Voltage Follower

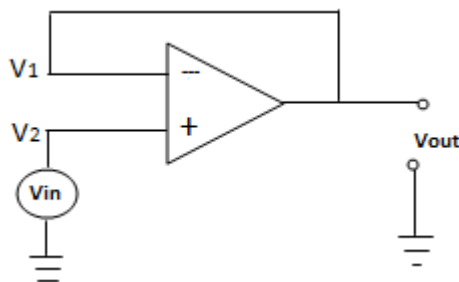


Figure 9. Voltage follower

A voltage follower is a circuit in which the output voltage v_{out} follows the input voltage, v_{in} . Circuit consists of non inverting amplifier where R_f is short circuited and R_1 is open circuited. Since input voltage is directly applied to non-inverting input terminal,

$$v_2 = v_{in} \quad (1)$$

Due to virtual short,

$$v_1 = v_2 \quad (2)$$

This implies that, $v_1 = v_2 = v_{in}$.

Inverting terminal is directly connected to output terminal,

$$\text{Hence, } v_{out} = v_1 \quad (3)$$

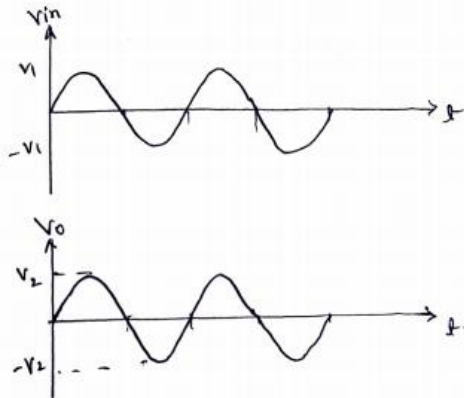
combining (1), (2) & (3)

$$\text{We have, } V_{out} = V_{in} \quad (4)$$

From (4), it is clear that output voltage follows input voltage.

Hence, the closed loop gain $A_f = \frac{v_{out}}{v_{in}} = 1$

Input/output waveforms



Inverting Op-Amp adder

Figure 10 shows the inverting configuration of op-amp with three inputs as v_1, v_2 and v_3 . A feedback resistor R_f and the input resistors R_1, R_2 and R_3 are connected in the circuit. Current through the resistors are i_1, i_2, i_3 and i_f . The output voltage of an inverting summer is equal to the negative sum of all the input times the gain of the circuit.

Here, $v_B = 0$, due to virtual short, $v_A = v_B = 0$

The currents in the circuit are,

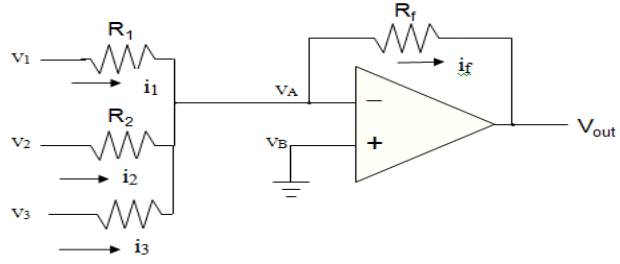


Figure 10. Inverting Summer

$$i_1 = \frac{v_1 - v_A}{R_1} = \frac{v_1}{R_1} \quad (1)$$

$$i_2 = \frac{v_2 - v_A}{R_2} = \frac{v_2}{R_2} \quad (2)$$

$$i_3 = \frac{v_3 - v_A}{R_3} = \frac{v_3}{R_3} \quad (3)$$

$$i_f = \frac{v_A - v_{out}}{R_f} = \frac{-v_{out}}{R_f} \quad (4)$$

Due to high impedance, the current flowing into inverting input terminal is zero.

Applying KCL, $i_f = i_1 + i_2 + i_3$ (5)

Sub (1), (2), (3) & (4) in (5)

$$\frac{-v_{out}}{R_f} = \frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3}$$

$$v_{out} = -\left[\frac{R_f v_1}{R_1} + \frac{R_f v_2}{R_2} + \frac{R_f v_3}{R_3}\right]$$

If $R_f = R_1 = R_2 = R_3$, then $v_{out} = -[v_1 + v_2 + v_3]$ (6)

Hence, output voltage is numerically equal to algebraic sum of output voltages $v_1, v_2, v_3 \dots v_n$.

Op-amp Integrator

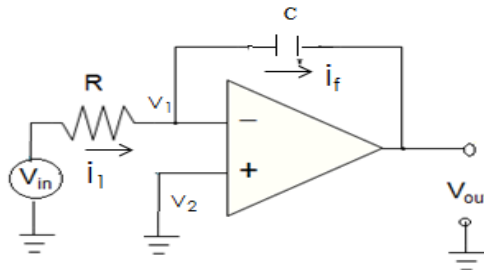


Figure 11. Integrator

In an integrator circuit, the output voltage is integration of the applied input voltage. When a step input voltage is applied, initially capacitor C is not charged and maximum current flows through resistor R . No current flows through amplifier input as $v_2 = 0$ and $v_1 = v_2$ due to virtual short. When the capacitor C begins to charge up to the input voltage, its impedance X_c increases slowly proportional to its rate of charge. The capacitor charges up to a rate determined by RC network. Ratio X_c/R increases and produces a ramp output till the capacitor is fully charged. During negative input voltage, capacitor discharges. Output voltage is inverted since v_{in} is applied to the inverting input terminal of the op-amp.

Voltage across the capacitor is,
$$v_c = \frac{Q}{C} \quad (1)$$

Due to virtual ground, $v_1 = 0$

Current through resistor R is

$$i_1 = \frac{v_{in} - v_1}{R}$$

$$i_1 = \frac{V_{in}}{R_{in}} \quad (2)$$

No current flows through inverting terminal of op-amp.

Current through C is i_f . Also voltage across the capacitor is

$$v_c = v_1 - v_{out}$$

$$\therefore v_c = -v_{out}, \quad (3)$$

we know that,
$$v_c = \frac{Q}{C} = -v_{out} \quad (4)$$

differentiating the equation (4), we get

$$\frac{1}{C} \frac{dQ}{dt} = - \frac{dv_{out}}{dt} \quad \text{or}$$

$$\frac{dQ}{dt} = -C \frac{dv_{out}}{dt}$$

The rate of change of charge is current i_f through the capacitor C and is given by

$$i_f = -C \cdot \frac{dv_{out}}{dt} \quad (5)$$

Equating (2) and (5) we get,

$$i_1 = i_f, \text{ i.e., we have}$$

$$\frac{V_{in}}{R} = -C \frac{dv_{out}}{dt}$$

Integrating on both sides, we get

$$\int_0^t \frac{v_{in}}{R} dt = -C \int \frac{dv_{out}}{dt} dt \quad \text{i.e.,}$$

$$\int_0^t \frac{v_{in}}{R} dt = -C v_{out}$$

$$\therefore v_{out} = -\frac{1}{RC} \int_0^t v_{in} dt \quad (6)$$

In Eq. 6, RC is known as time constant. Negative sign indicates that there is a phase shift of 180° between input and output signals.

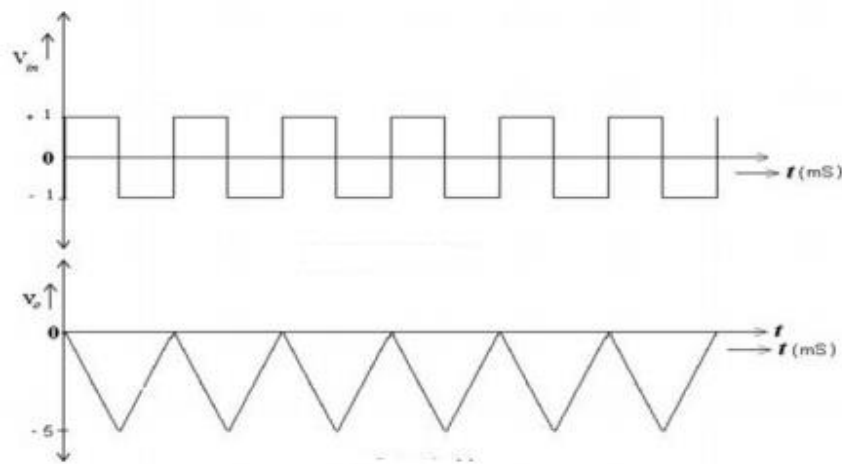


Figure: Input/output waveforms of Op-amp integrator

Op-amp Differentiator

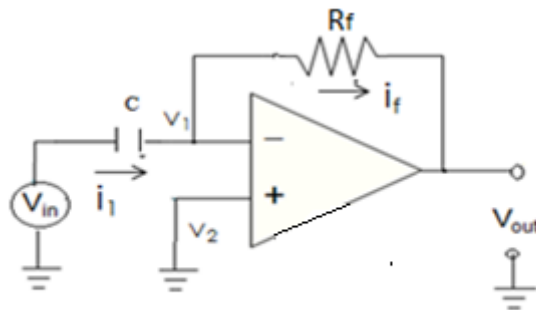


Figure 12.a Differentiator

The circuit which produces differentiation of the applied input voltage at its output is called as differentiator. The input signal is applied to a capacitor. The capacitor blocks DC & allows only AC voltage to pass. The capacitor will charge to the applied input voltage. Non inverting terminal is at ground potential and hence $v_1 = 0$. According to virtual short concept, $v_2 = 0$. Hence, no current flows into the input terminals of the op-amp.

As capacitor charges, voltage across it is $v_c = \frac{Q}{C}$ (1)

$$v_c = \frac{Q}{C} = v_{in} - v_1, \quad (2)$$

Differentiating (2) on both side, we get

$$\frac{dQ}{dt} \frac{1}{C} = \frac{d(v_{in} - v_1)}{dt}$$

Due to virtual ground, $v_1 = 0$

$$\therefore \frac{dQ}{dt} = C \frac{dv_{in}}{dt}$$

Rate of change of charge is current i_1 through the capacitor.

$$i_1 = C \frac{dv_{in}}{dt} \quad (3)$$

Current through the resistor R_f is

$$i_f = \frac{v_1 - v_{out}}{R_f}$$

$$i_f = \frac{-v_{out}}{R_f} \quad (4)$$

Equating (3) and (4), we have

$$i_i = i_f$$

$$C \frac{dv_{in}}{dt} = \frac{-v_{out}}{R_f}$$

$$v_{out} = -CR_f \frac{dv_{in}}{dt} \quad (5)$$

Negative sign indicates that there is a phase shift of 180° between input & output signal. CR_f is the time constant of the differentiator.

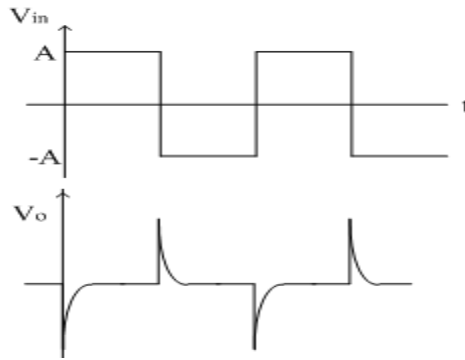


Figure 12.b: Input/ output waveforms of Op-amp differentiator

Op-Amp Comparators

A comparator is a circuit which compares a signal voltage applied at one input of an op-amp with a known reference voltage at the other input, and produce either a high or low output voltage, depending on which input is higher. Op-amp is used as a comparator in its open loop configuration.

Non-inverting Comparator: Here, the reference voltage is applied at the input of the op-amp & time varying signal V_{in} is applied at the non-inverting input of the op-amp. Since, time varying signal is applied at the non-inverting input of op-amp, the circuit is called as non-inverting comparator.

Case 1: with positive V_{ref} : The open loop gain of op-amp is very large. With a small input voltage, the op-amp will reach saturation and the output voltage will be either $+V_{sat}$ or $-V_{sat}$. When V_{in} is less than V_{ref} , the output V_{out} is at $-V_{sat}$ ($\cong -V_{EE}$). This is because voltage at the inverting input is higher than that at the non-inverting input. When V_{in} is greater than V_{ref} , the non-inverting input becomes positive w.r.t inverting input and V_{out} goes to $+V_{sat}$ ($\cong V_{CC}$). So the output is at $+V_{sat}$ as shown in figure.

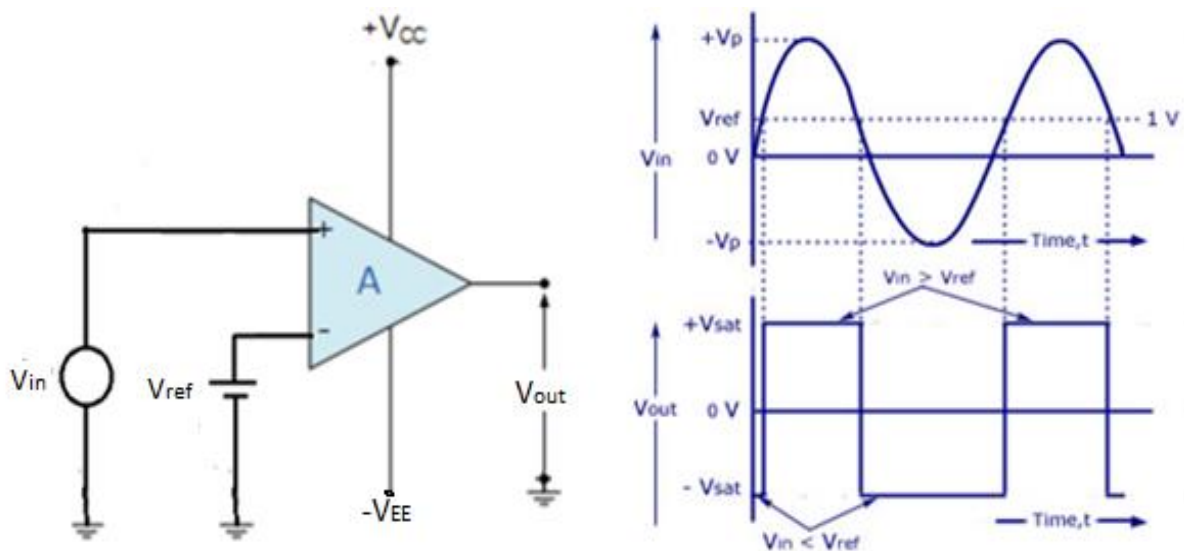


Figure 13. Non inverting comparator with $+V_{ref}$ voltage

Case 2: with negative V_{ref} : When V_{in} is greater than V_{ref} , the non-inverting input becomes positive w.r.t inverting input and hence the output V_{out} goes to $+V_{sat}$ ($\cong +V_{CC}$). When V_{in} is less than V_{ref} , voltage at inverting terminal is greater than voltage level at the non inverting terminal. Hence the output voltage V_{out} is at $-V_{sat}$ ($\cong -V_{EE}$).

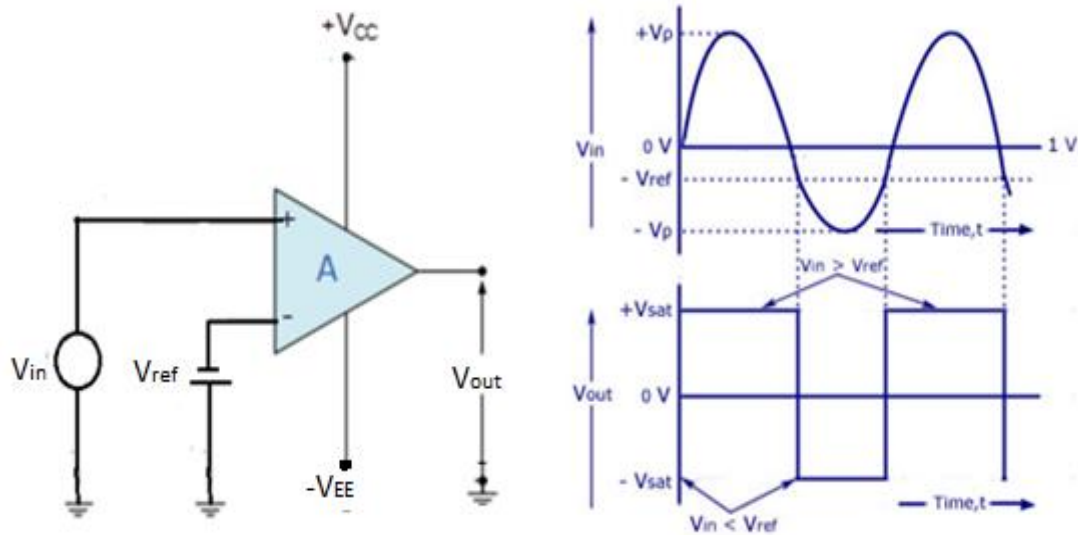


Figure 14. Non inverting comparator with $-V_{ref}$ voltage

Inverting Comparator: Here, the reference voltage V_{ref} is applied at the non-inverting input of the op-amp and time varying signal V_{in} is applied at the inverting input of the op-amp. Since, time varying signal is applied at the inverting input of op-amp, the circuit is called as inverting comparator.

Case 1: with positive V_{ref} : In this, the reference voltage V_{ref} is applied to non-inverting input and time varying signal voltage V_{in} is applied to inverting input. When V_{in} is less than V_{ref} , the non-inverting input becomes positive w.r.t inverting input and hence the output voltage V_{out} goes to $+V_{sat}$ ($\cong +V_{CC}$). When V_{in} is greater than V_{ref} , the inverting input becomes positive w.r.t non inverting input and hence the output V_{out} is at $-V_{sat}$ ($\cong -V_{EE}$). Since the open loop gain of op-amp is very large with a small input voltage, the op-amp will drive into saturation and the output voltage will be either $+V_{sat}$ or $-V_{sat}$.

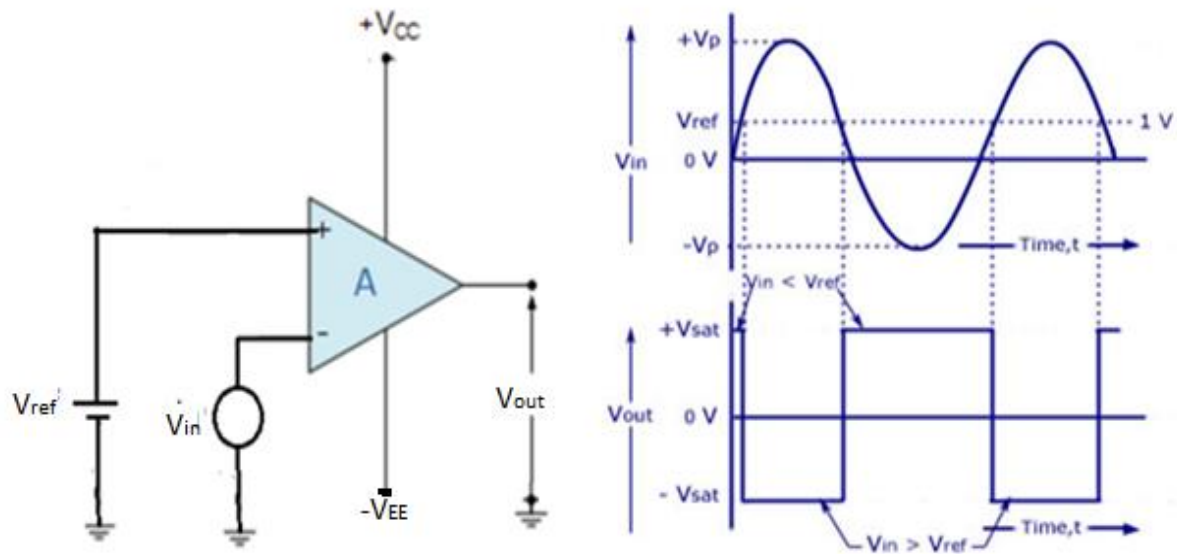


Figure 15. Inverting comparator with $+V_{ref}$ voltage

Case 2: with negative V_{ref} : When V_{in} is greater than V_{ref} , the non-inverting input becomes negative w.r.t inverting input and hence the output voltage V_{out} goes to $-V_{sat}$ ($\cong -V_{EE}$). When V_{in} is less than V_{ref} , the output voltage V_{out} is at $+V_{sat}$ ($\cong +V_{CC}$) because the voltage at the inverting input is less than that of non-inverting input.

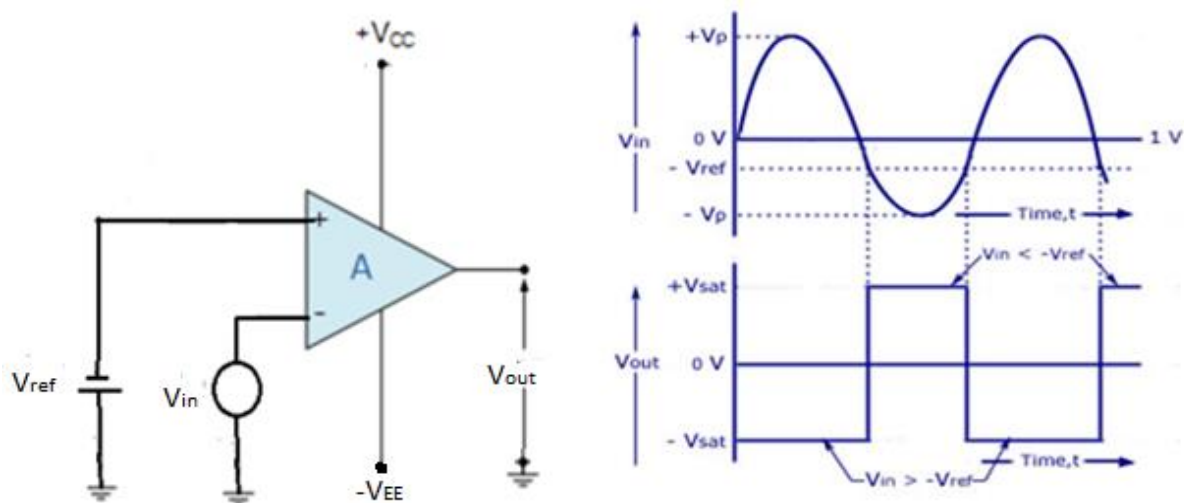


Figure 16. Inverting comparator with $-V_{ref}$ voltage

IC 555 TIMER AS AN OSCILLATOR

IC 555 is a versatile linear IC introduced during 1970. It is a Monolithic IC used for many applications such as astable multivibrator, pulse detector automatic battery charger etc.

Features of IC 555 timer: It can produce accurate timer delays or oscillations ranging from few microsecond to several hours.

Two modes of operation are: Monostable and Astable.

Supply voltage range: 4.5V to 18V.

Current: 200 mA.

It is an 8 pin IC package.

Pin diagram:

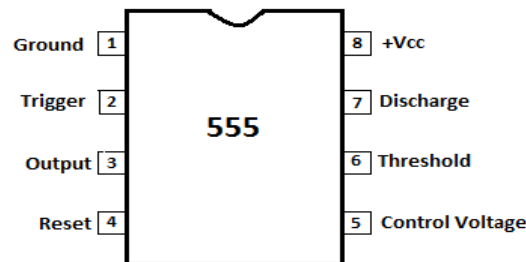


Figure 17. Pin diagram

Pin 1: It is a ground pin. Voltages are measured with respect to 1.

Pin 2 and 6: Trigger and threshold pins. A chain of three resistors of equal values (R) as $5K\Omega$ are connected through V_{CC} . Hence a reference voltage of $2/3 V_{CC}$ appears at comparator 2 and $1/3 V_{CC}$ at comparator 1.

Pin 3: It is an output pin, output signal is collected.

Pin 4: It is a Reset pin. This pin helps for capacitor discharge.

Pin 5: It is a Control voltage pin that is connected to $2/3 V_{CC}$ to change the levels of voltage for other applications.

Pin 7: It is a discharge pin that is connected to collector of the transistor. It gives a path for the capacitor to discharge based on the output Q of RS flip flop.

Pin 8: It is a power supply pin connected to $+V_{CC}$.

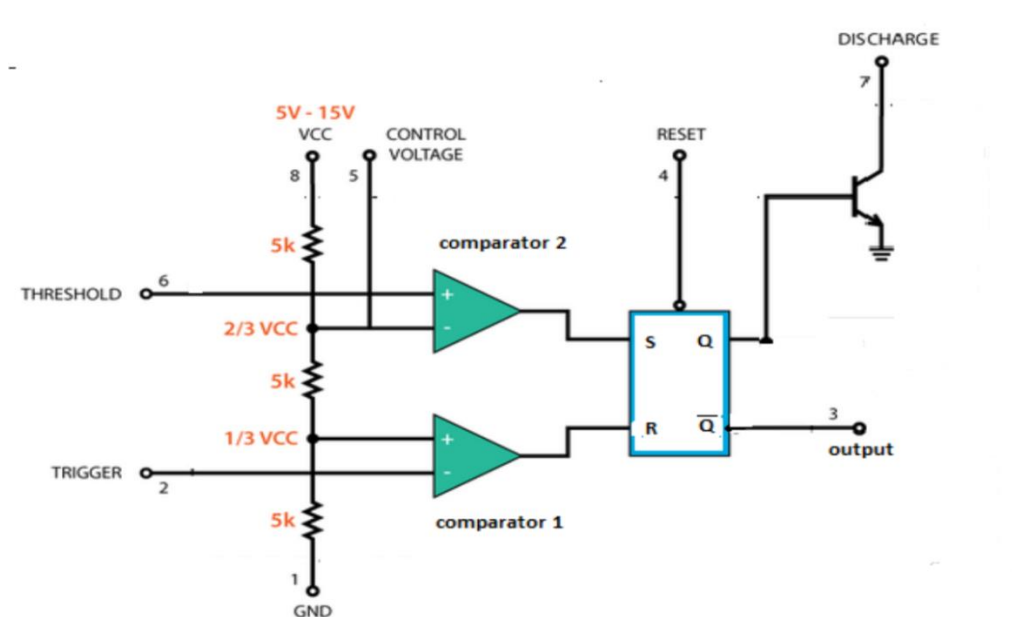


Figure 18. Internal diagram of IC 555

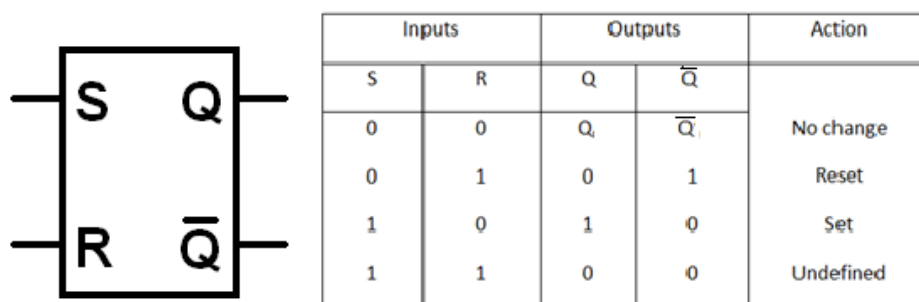


Figure 19. SR- flip flop and truth table

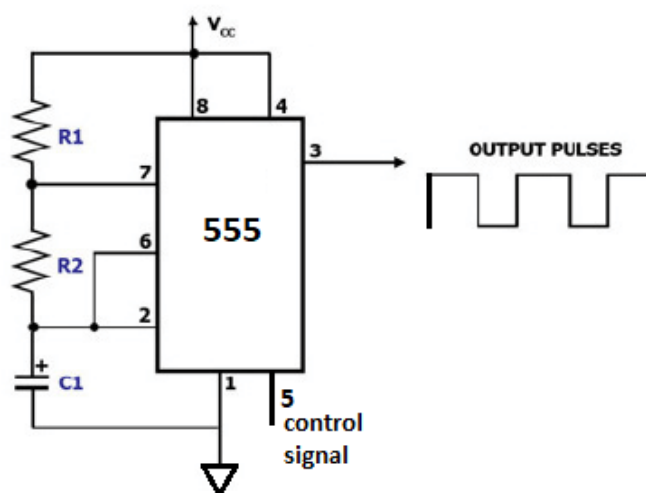


Figure 20. Circuit diagram

IC 555 as an oscillator:

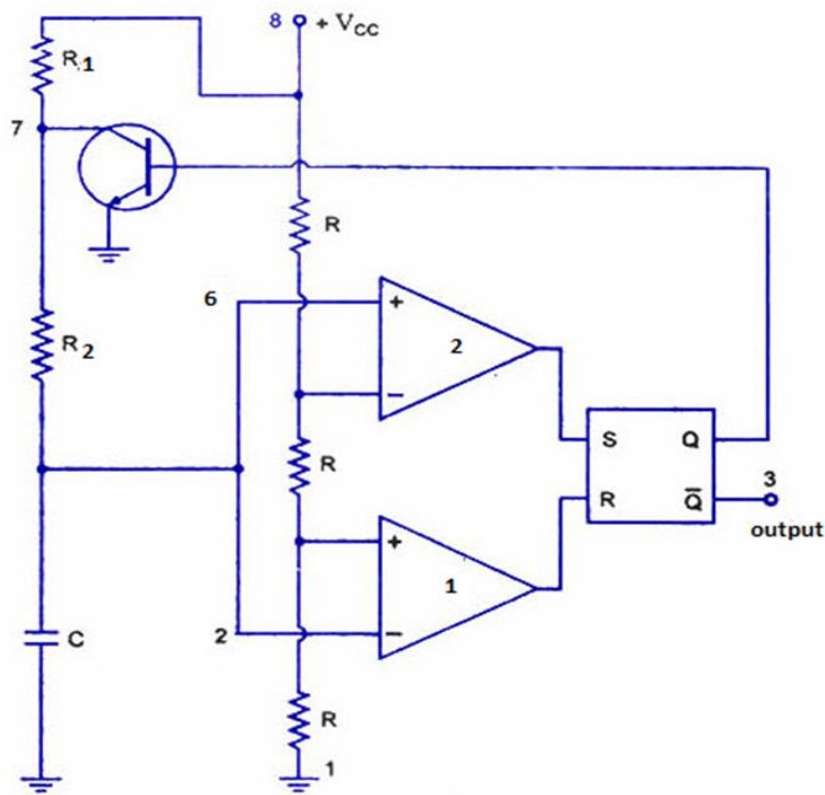


Figure 21. Internal schematic of IC 555 timer

Operation of IC 555 timer as an oscillator (astable mode):

WORKING:

Initially when the power supply is ON, capacitor C starts charging through resistors R_1 and R_2 to reach the peak value $+V_{CC}$.

As C charges in the voltage level at Pin 2 (Trigger) is less than $1/3V_{CC}$, Comparator 1 output goes high (1). Therefore $R = 1$, $S = 0$. Hence $Q = 0$ and output at pin 3 is high, $\bar{Q} = 1$.

The capacitor C continues to charge beyond $1/3V_{CC}$ and when the voltage is less than $2/3V_{CC}$, both comparator outputs are zero and there is no change in the output at pin number 3.

When the voltage at pin number 6 threshold goes slightly above $2/3V_{CC}$, output of Comparator 2 goes high (1). This sets flip-flop to $S=1$ and $R=0$. Hence $Q = 1$ and output at Pin 3 goes low (0).

Since $Q = 1$, there is a base drive to the transistor T and it is ON. Now the collector of T is connected to discharge pin 7. The capacitor which had charged to $2/3V_{CC}$ now starts discharging through R_2 to ground.

When the voltage across the discharging capacitor is between $2/3V_{CC}$ and $1/3V_{CC}$, the comparator output is zero and there is no change in output at pin 3. Hence the output $\bar{Q}$ will have the previous state.

When the voltage level at pin 2 goes slightly below $1/3V_{CC}$, Comparator 1 output goes high. Therefore $R = 1$ and $S = 0$. Hence $Q = 0$ and output at pin 3 is high (1).

When $Q = 0$, there is no base drive to T. Hence transistor T is cut-off. i.e. Collector gets disconnected from discharge pin 7 thereby allowing capacitor C to continuously charge again from $1/3V_{CC}$ through R_1 and R_2 . This process repeats & generates square wave or oscillations at the output pin 3.

Time taken by the capacitor to charge is

$$T_{ON} = 0.693(R_1 + R_2)C$$

Time taken by the capacitor to discharge is

$$T_{OFF} = 0.693 R_2 C$$

Frequency of oscillation is, $f = \frac{1}{T}$

$$\text{Total time } T = T_{ON} + T_{OFF}$$

$$T = 0.693(R_1 + R_2)C + 0.693 R_2 C$$

$$\therefore f = \frac{1}{T} = \frac{1.44}{(R_1 + 2R_2)C}$$

$$\therefore \text{Duty cycle} = \frac{T_{ON}}{T_{ON} + T_{OFF}} = \frac{R_1 + R_2}{R_1 + 2R_2} = \frac{T_{ON}}{T}$$

Table: Working logic of IC555 timer

VOLTAGE LEVELS	COMP1	COMP2	R	S	Q	$\bar{Q}$
$V_C < \frac{1}{3}V_{CC}$	1	0	1	0	0	1
$\frac{1}{3}V_{CC} < V_C < \frac{2}{3}V_{CC}$	0	0	0	0	0	1

$V_C > \frac{2}{3}V_{CC}$	0	1	0	1	1	0
$\frac{1}{3}V_{CC} < V_C < \frac{2}{3}V_{CC}$	0	0	0	0	1	0
$V_C < \frac{1}{3}V_{CC}$	1	0	1	0	0	1

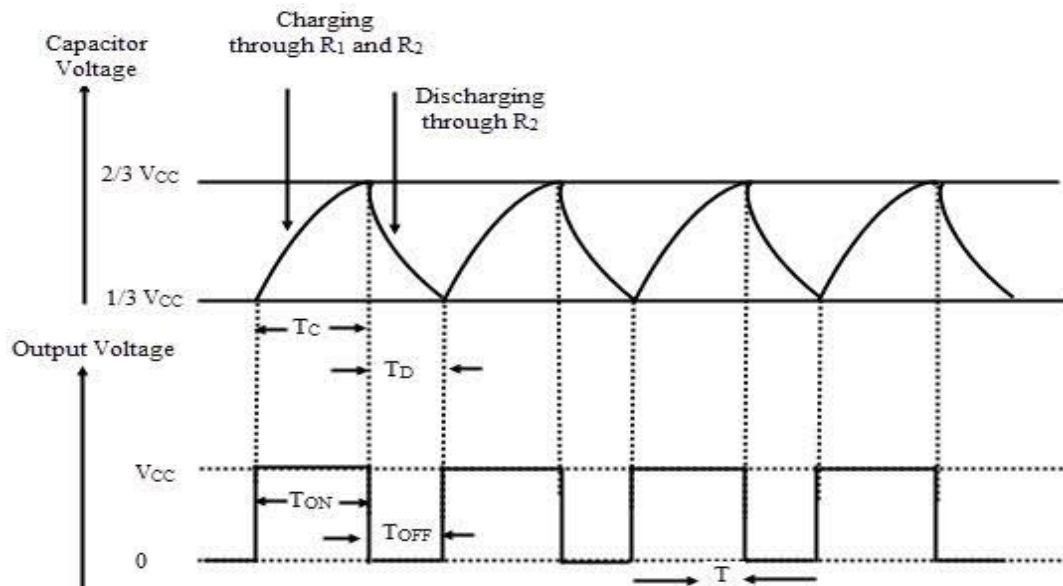


Figure 22: Voltage across capacitor at pin 2 & Output voltage at pin 3

PROBLEMS ON IC 555 TIMER:

- 1) For an IC 555 timer, $T_{ON} = 3\text{sec}$, $T_{OFF} = 1\text{sec}$ and $C = 10\mu\text{F}$. Calculate the Value of R_1 and R_2 .

Solution:

$$\text{Given: } T_{ON} = 3\text{sec}, T_{OFF} = 1\text{sec}, C = 10\mu\text{F}$$

Solution:

$$T_{OFF} = 0.693 R_2 C$$

$$1 = 0.693 \times R_2 \times 10 \times 10^{-6}$$

$$R_2 = 144 \text{ K}\Omega$$

$$T_{ON} = 0.693(R_1 + R_2)C$$

Therefore $R_1 = 288.6 \text{ K}\Omega$

- 2) For an IC 555 timer, given $D = 75\%$, $f = 1 \text{ KHz}$, $R_2 = 3.6 \text{ k}\Omega$, $C = 0.1 \mu\text{F}$. Calculate R_1 .
Solution:

$$\text{Given: } D = 75\% \Rightarrow 0.75$$

$$f = 1 \text{ kHz}, R_2 = 3.6 \text{ k}\Omega, C = 0.1 \mu\text{F}$$

$$D = \frac{T_{ON}}{T}$$

$$T = \frac{1}{f} \Rightarrow 1 \text{ m sec}$$

$$T_{ON} = D \times T$$

$$T_{ON} = 0.75 \times 10^{-3} \text{ sec} = 750 \mu\text{sec}$$

$$T_{OFF} = 0.693 R_2 C$$

$$T_{OFF} = 249 \mu\text{sec}$$

Since $D > 50\%$, $T_{ON} > T_{OFF}$

$$T_{ON} = 0.693(R_1 + 3600)0.1 \times 10^{-6}$$

Therefore $R_1 = 7.215 \text{ k}\Omega$

74XX series based fixed IC voltage regulator

A voltage regulator is one of the most widely used electronic circuitry in any device. A regulated voltage (without fluctuations & noise levels) is very important for the smooth functioning of many digital electronic devices. A common case is with micro controllers, where a smooth regulated input voltage must be supplied for the micro controller to function smoothly. Integrated Circuits regulators are available with fixed positive or negative output voltages or variable negative or positive output voltages.

Figure shows the functional diagram for 74XX series. Control element (pass transistor) in series with load between input and output. A reference voltage V_{ref} drives the non-inverting input of an amplifier. The resistor R_1 and R_2 sense a change in the output voltage and provide a feedback voltage. Output **sample circuit** senses a change in output voltage. The error detector compares the feedback voltage with a Zener diode reference voltage. **Error detector** compares sample voltage with reference voltage causes control element to compensate in order to maintain a constant output voltage. The resulting difference voltage causes the transistor Q_1 controls the conduction to compensate the variation of the output voltage. The block diagram of the IC voltage regulator is as shown in Fig.23.

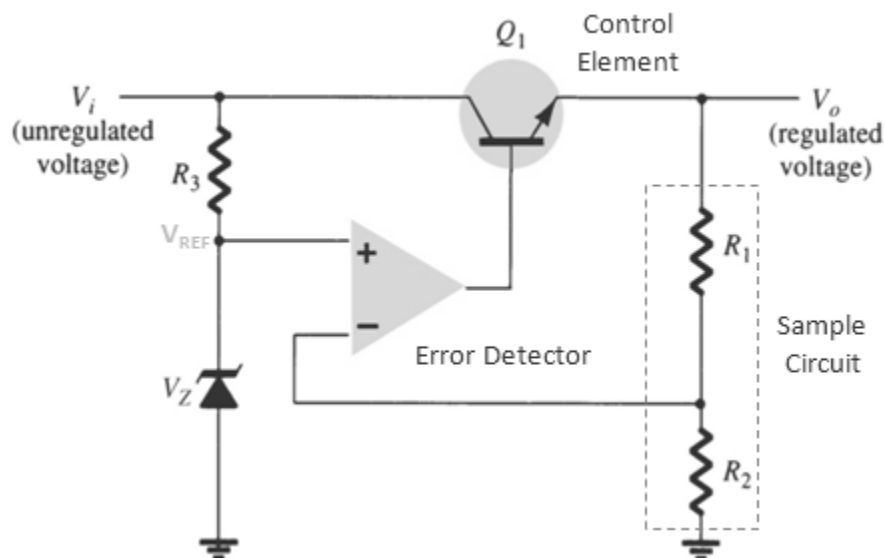


Fig.23: Circuit diagram of 74XX series based fixed IC voltage regulator

The output voltage will be maintained at a constant value of:

$$v_o = \left[1 + \frac{R_1}{R_2} \right] v_z \text{ volt}$$

~~get a positive half cycle at the output for the negative half cycle of the input. Hence 180° phase shift is observed between the input and output voltages for a CE amplifier.~~

Feedback amplifiers

The purpose of an amplifier is to amplify (increase the magnitude) of the input signal without changing any of its characteristics except the amplitude. Feedback plays an important role in electronic circuits. The performance of an amplifier can be improved by using feedback network which connects the output to the input of an amplifier. Block diagram of a simple feedback amplifier is as shown in Fig. 24

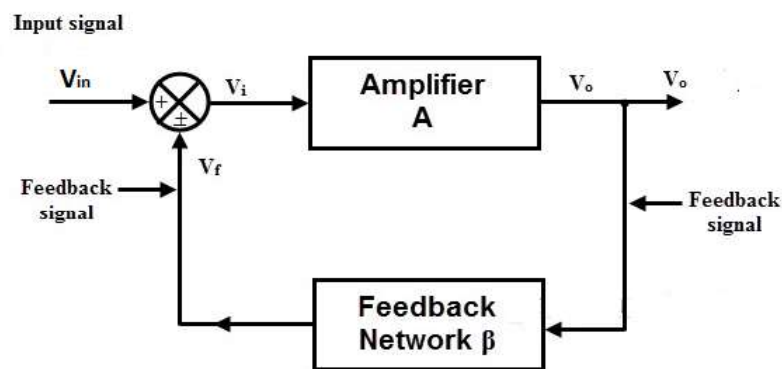


Fig. 24 Block diagram of a simple feedback amplifier

The input signal, V_{in} is applied to a mixer network in which it is combined with a feedback signal, V_f . The difference of these signals, V_i is the input to the amplifier. A portion of the amplifier output is sampled & connected to the input side through a feedback network (β).

If the feedback signal, V_f is in phase with the input signal, V_{in} then the feedback is called as positive feedback. If the feedback signal, V_f is out of phase with the input signal, V_{in} then the feedback is called as negative feedback.

Negative feedback leads to amplification whereas positive feedback leads to oscillations.

Concept of negative feedback:

Consider a block diagram of a negative feedback amplifier as shown in Fig 25 In the figure, gain 'A' is the gain of the amplifier without feedback. Its value is always greater than '1' as V_o is always greater than V_i .

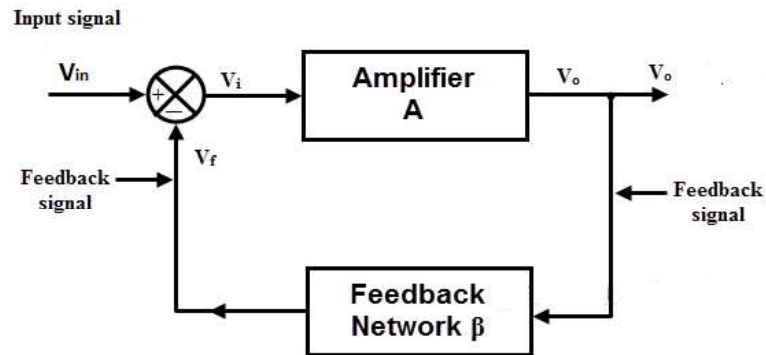


Fig.25: Negative feedback amplifier

Here, $A = \frac{V_o}{V_i}$ is the gain of the basic amplifier (without feedback)

‘A’ is known as **open loop gain**

$A_f = \frac{V_o}{V_{in}}$ is the overall gain of the amplifier (with feedback)

A_f is also known as **closed loop gain**

here, V_f is the feedback signal and β is the feedback factor.

Overall gain or gain with feedback is

$$A_f = \frac{V_o}{V_{in}} = \frac{A}{1 + A\beta},$$

with $|1 + A\beta| > 1$ and $A_f < A$

Thus the negative feedback reduces the gain by a factor of $(1 + A\beta)$. Negative feedback is also called as degenerative feedback.

Advantages of negative feedback:

- 1) Stabilized gain
- 2) Higher input impedance

- 3) Lower output impedance
- 4) Increased bandwidth
- 5) Reduction in noise.

Voltage Series Negative Feedback:

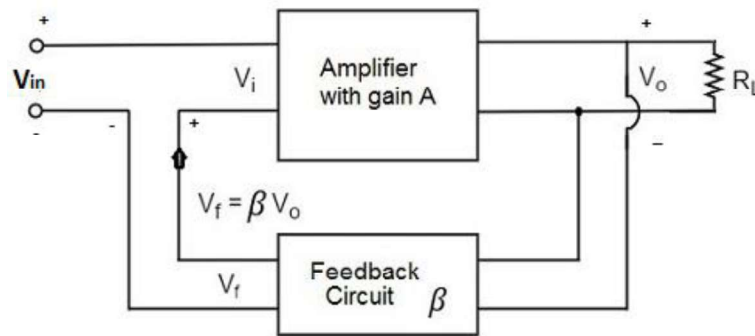


Fig. 26: Voltage Series Negative Feedback

The Fig. 26 shows the block diagram representation of a voltage series negative feedback amplifier. Here, the output voltage, V_o is sampled (tapped) at the output and fed back in series with the input. It is a voltage amplifier.

Assume a sinusoidal input signal, V_{in} of the amplifier which is an inverting amplifier. i.e., output has 180° phase shift w.r.t input. A part of the output voltage is sampled using a feedback network and is fed back to the input. Let 'A' be the gain of the amplifier and β be the feedback factor.

$$\text{Then } V_f = \beta V_o \text{ or } \beta = \frac{V_f}{V_o}.$$

The value of β lies between 0 & 1.

When feedback factor β is 0, it means that no feedback is applied. When β is 1, whole of the output voltage V_o is applied as a feedback voltage.

$$\text{i.e., } V_f = V_o$$

Since the amplifier is of inverting type, V_o & V_{in} will be out of phase by 180° . Thus, V_{in} & V_f will be out of phase as in the block diagram. Hence, the net input voltage, V_i to the

amplifier is the difference between V_{in} & V_f . Therefore, this type of feedback is known as negative feedback.

$$V_i = V_{in} - V_f \quad \text{or}$$

$$V_{in} = V_i + V_f$$

But the gain with feedback is

$$A_f = \frac{V_o}{V_{in}}$$

$$A_f = \frac{V_o}{V_i + V_f} = \frac{V_o}{V_i + \beta V_o},$$

but, $A = \frac{V_o}{V_i}$ simplifying, we have,

$$A_f = \frac{A}{1 + A\beta}$$

This is a general representation of gain.

A_f is called as closed loop gain, since the gain is measured with feedback.

In a negative feedback system, the feedback sample V_f is subtracted from the signal input V_{in} .

$$|A_f| < |A|, \text{ also } |1 + A\beta| > 1$$

i.e., gain with feedback is always much less than gain without feedback.

Hence A_f is small.

Problem:

An amplifier has an open loop voltage gain of 1000. If the feedback factor is 10 %, find the closed loop voltage gain.

Solution: $A=1000$, $\beta=10\%= 10/100 = 0.1$

Closed loop gain is
$$A_f = \frac{A}{1 + A\beta} = \frac{1000}{1 + 1000(0.1)} = 9.9$$

Concept of positive feedback:

When the input signal and part of the output signal is fed back in phase, the feedback is called as positive feedback. Positive feedback results in oscillations.

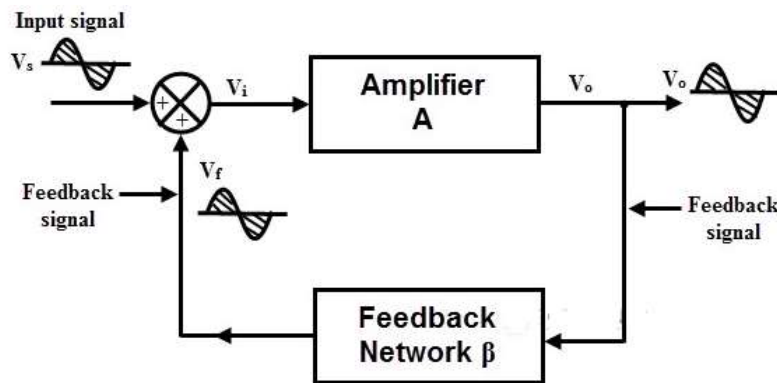


Fig.27: Positive Feedback Amplifier

Consider an inverting amplifier with voltage gain, A as in Fig.27. Assume that sinusoidal input signal, V_{in} is applied to the block. As the amplifier is of inverting type, the output voltage, V_o is out of phase by 180° w.r.t input signal, V_{in} . A part of V_o is fed back to the input with the help of the feedback network. The amount of V_o that has to be fed back is decided by the feedback factor, β . The feedback network introduces a phase shift of 180° to V_o . Phase of the feedback signal is same as that of the input signal applied. Hence the feedback is called as positive feedback.

Expression for gain A_f with feedback:

For the overall block, the input supply is V_{in} & output is V_o . The ratio of output voltage, V_o to the input voltage, V_{in} with corresponding effect of feedback is called as closed loop gain of the network or gain with feedback denoted as A_f .

$$A_f = \frac{V_o}{V_{in}}$$

Since the feedback is positive & voltage V_f is added to V_{in} to generate the input signal V_i to the amplifier, from the block diagram, we can write,

$$V_i = V_{in} + V_f$$

V_f depends on β of the feedback network.

$$V_f = \beta V_o$$

Substituting V_f in V_i

$$V_i = V_{in} + \beta V_o$$

$$\therefore V_{in} = V_i - \beta V_o$$

$$A_f = \frac{V_o}{V_{in}} = \frac{V_o}{V_i - \beta V_o} = \frac{V_o / V_i}{1 - \beta V_o / V_i}$$

$$\therefore A_f = \frac{A}{1 - A\beta}$$

It can be observed as how gain with feedback varies with varying β as in table below:

A	β	A_f
20	0.005	22.22
20	0.04	100
20	0.045	200
20	0.05	∞

We observe that gain with feedback increases as the amount of positive feedback increases & reaches infinity. This indicates that the circuit can produce output without external input ($V_{in} = 0$) signal just by feeding the part of the output as its own input. Hence, the output voltage cannot be infinity but gets driven to oscillations. β value must be less than 1. To start

with oscillations, $A\beta > 1$ must be set. But the circuit adjusts itself to get $A\beta = 1$ to produce sinusoidal oscillations while working as an oscillator. An oscillator is an amplifier that uses positive feedback & without any external input signal, generates an output waveform at a desired frequency.

Barkhausen's criterion for oscillations:

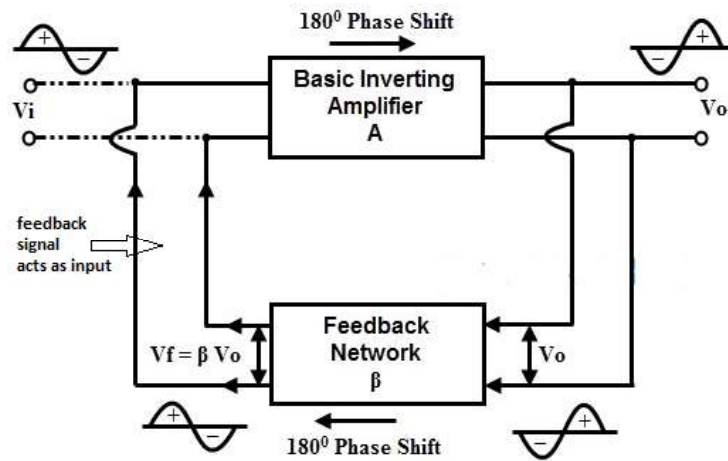


Fig. 28: Block diagram of an Oscillator

The two conditions required for the circuit to work as an oscillator are called as Barkhausen's criteria for sustained oscillations

Barkhausen's criteria states that

- 1) Total phase shift around the loop as the signal proceeds from input through the amplifier, feedback network and back to input again completing a loop is 0° or 360° .
- 2) The magnitude of the product of the open loop gain A & the feedback factor is $|A\beta| = 1$

Satisfying these conditions, the circuit works as an oscillator producing sustained oscillations.

Inverting amplifier has a gain of A with phase shift of 180° . Let the input, V_{in} applied to the amplifier be derived from its output using a feedback network. The network has feedback factor, β which is less than 1. The amplifier produces a phase shift of 180° . The output voltage fed back using feedback network must be in phase with V_{in} . Thus the feedback

network must introduce a phase shift of 180° so that the total phase shift is $0^\circ / 360^\circ$ so that V_{in} is in phase with V_f .

Fig. 28 describes the block diagram of an oscillator. Let V_{in} be a fictitious voltage applied as input to the amplifier.

$$V_o = AV_{in} \quad \dots\dots\dots(1)$$

The feedback factor β divides the amount of feedback to be given to the input.

$$V_f = \beta V_o \quad \dots\dots\dots(2)$$

Substituting Eq. 1 in 2, we get

$$\therefore V_f = A\beta V_{in} \quad \dots\dots\dots(3)$$

For the oscillator, the feedback voltage V_f should drive the amplifier & hence V_f must act as V_{in} . From Eq. 3, we conclude that V_f is sufficient to act as V_{in} when

$$|A\beta| = 1$$

$$\text{ie, } V_f = A\beta V_{in}$$

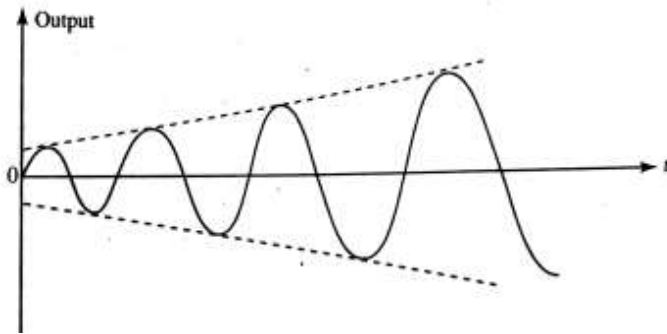
$$\text{or } V_f = V_{in}$$

Phase of V_f should be same as V_{in} . Amplifier introduces the phase shift of 180° & feedback network with 180° . This ensures positive feedback. Hence total phase shift around loop is 360° .

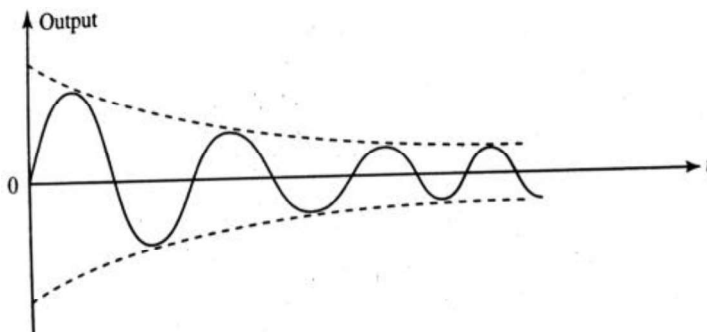
Under this condition, V_f drives the circuit without any external input, hence working as an oscillator.

Cases:

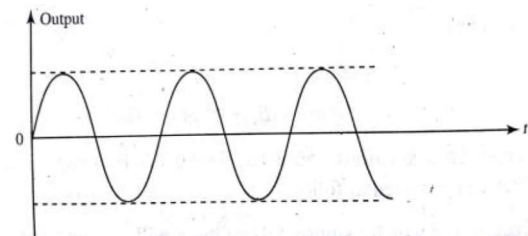
Case 1: $|A\beta| > 1$, oscillations are of growing type



Case 2: $|A\beta| < 1$, oscillations are of decaying type



Case 3: $|A\beta|=1$, oscillations with constant frequency and amplitude called as sustained oscillations



~~Oscillators:~~

~~The circuit which generates waveforms without any input given to it is called an oscillator. It is dc to ac converter.~~

~~Classification of oscillators:~~

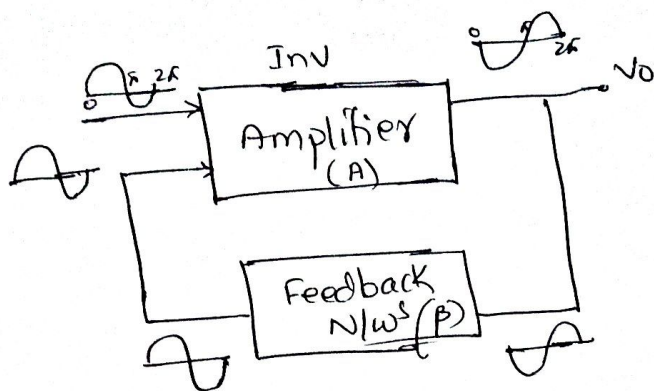
~~We are going to read mainly about three types of oscillators:~~

- ~~(1) RC Phase Shift Oscillator~~
- ~~(2) Hartley Oscillator~~
- ~~(3) Colpitt's Oscillator~~

Op-amp oscillators

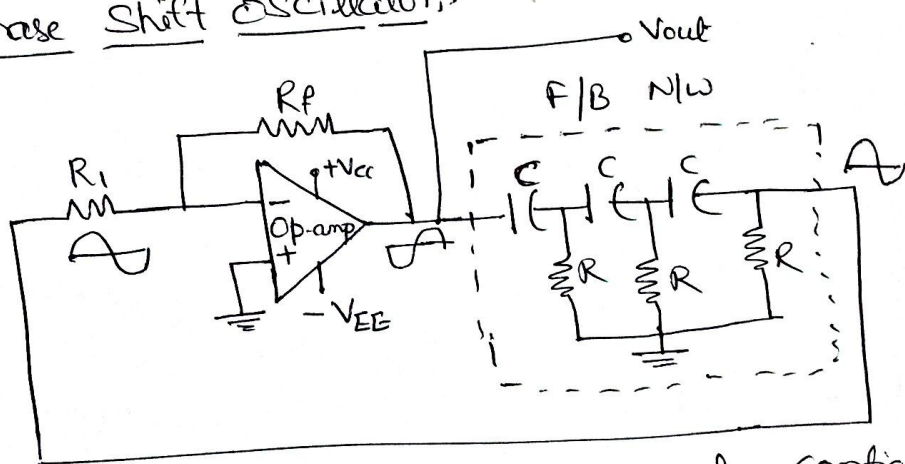
①

Phase Shift Oscillators:-



- * Condition for oscillation: $|A\beta| > 1$ & total phase shift must be $360^\circ / 0^\circ$
- * Feedback network must provide a phase shift of 180° to make the positive feedback.

RC phase shift oscillator:-



- * Op-amp is used in inverting mode configuration.
- * Each RC n/w @ f.b provides a 60° phase shift causing a total 180° phase shift.

- * Frequency of oscillation,

$$f = \frac{1}{2\pi RC\sqrt{6}} \text{ Hz}$$

- * Other name of above RC n/w is Ladder.

* All values of R & C are same.

* For a particular frequency each RC section produces a 60° phase shift.

$$\phi = \tan^{-1} \left(\frac{X_C}{R} \right) \quad |Z| = \sqrt{R^2 + X_C^2}$$

* By adjusting R & C, ϕ can be changed

* The Op-amp is configured as an inverting amplifier

* O/p of amplifier is given to the F/B N/w.

* O/p of F/B N/w drives the amplifier.

* Total phase shift around the closed loop is,
 $180^\circ + 180^\circ = 360^\circ$ (+ve F/B)

Advantages

* Simple

* Generates signal in AF range

* Fixed frequency osc.

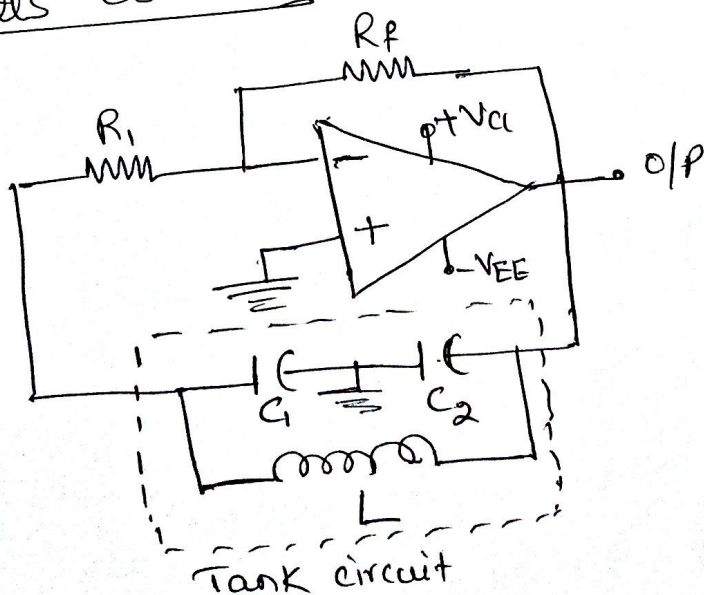
Disadvantages

* If R & C is changed, F/B conditions will change.

* Oscillator conditions are not satisfied

Tuned Oscillator circuit

Colpitts oscillator



$$f_0 = \frac{1}{2\pi \sqrt{L C_{eq}}} \text{ Hz}$$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

* Tank circuit provides a 180° phase shift.

* $X_C = \frac{1}{\omega C}$ $X_1 = X_2 = C$ $X_3 = L$

* Feedback factor, $\beta = \frac{C_2}{C_1}$; $\frac{XC_2}{XC_1} = \frac{\frac{1}{\omega C_2}}{\frac{1}{\omega C_1}} \Rightarrow \frac{C_1}{C_2}$

* Condition for sustained oscillation is,

$$|A_v \beta| > 1 \Rightarrow |A_v| > \frac{1}{\beta}$$

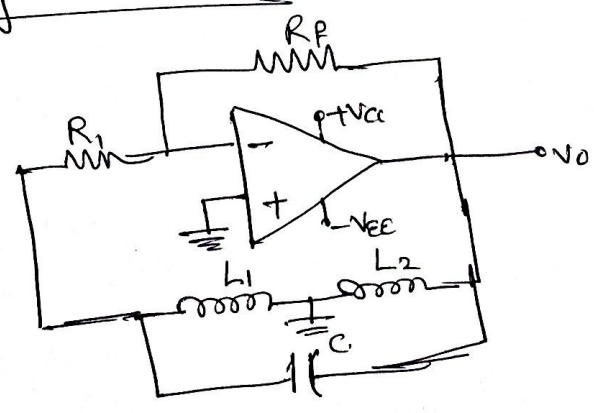
$$|A_v| > \frac{C_1}{C_2}$$

* Used as local oscillator in super heterodyne receivers.

* When supply voltages are applied oscillation currents are set up in the tank circuit which produces voltage a/c C_1 & C_2 .

* C_1 & C_2 forms capacitive divider n/w.

Hartley oscillator



$$f = \frac{1}{2\pi \sqrt{L_{eq} C}} \text{ Hz}$$

$$L_{eq} = L_1 + L_2 + 2M$$

$M \rightarrow$ mutual inductance b/w L_1 & L_2

* $\beta = \frac{XL_1}{XL_2} = \frac{\omega L_1}{\omega L_2} = \frac{L_1}{L_2}$

* Condition for sustained oscillation, $|A_v \beta| > 1$

$$|A_v| > \frac{1}{\beta} \Rightarrow |A_v| > \frac{L_2}{L_1}$$